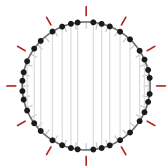
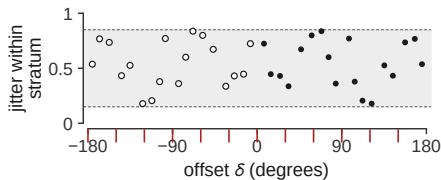


## A Fixed nonuniform offset design

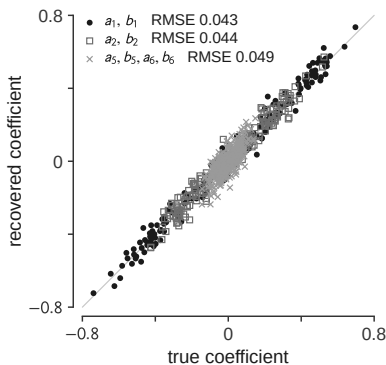


36 fixed offsets, mirror pairs joined  
regular 12-point design (outer ticks)



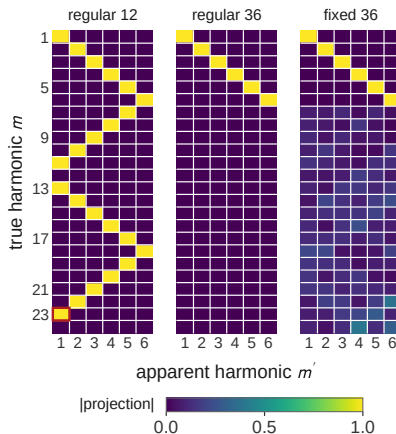
36 offsets = 18 exact mirror pairs; no anchor at  $\delta=0$ .  
Stratified jitter in  $\pi/18$  strata, seed 2026102300.  
Gaps 6.08–14.47°; jitter within [0.15, 0.85].

## D Recovery on measured offsets



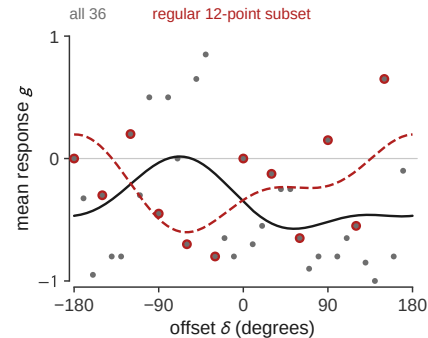
90 mixtures, 24 responses per offset, order-6 fit on the 36 fixed offsets, with an out-of-band  $m=23$  term at the measured amplitude 0.242.

## B Aliasing onto low-order modes



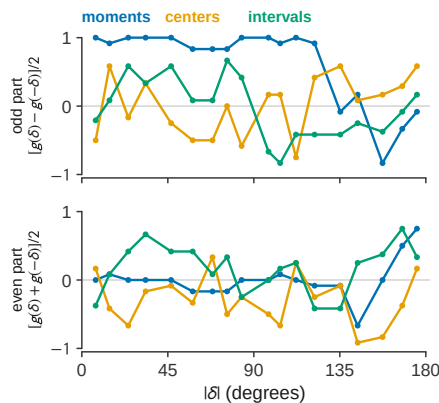
Regular 12-point sampling maps  $m=23$  onto  $m'=1$  with unit gain (red cell).  
That harmonic carries amplitude 0.242 in the narrow field; the fixed design leaks 0.075 of it. A regular 36-point grid is clean to  $m=29$ , not beyond.

## C Measured sign change



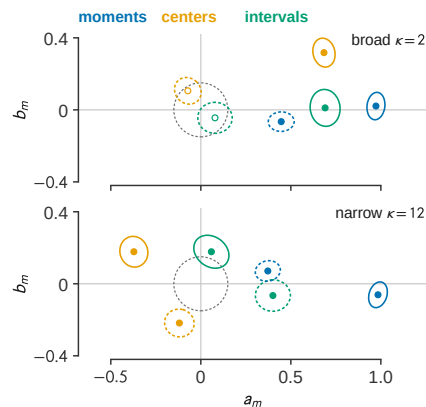
centers, narrow field  $\kappa=12$ ; 40 per offset.  
36-point grid, black fit:  $a_1 = -0.242$   
12-point subset, red fit:  $a_1 = +0.134$   
other subset phases:  $-0.459, -0.400$

## E Mirror-pair balance



Unimodal  $\kappa=9$ , block 1, 18 mirror pairs.  
The design supplies both halves of every pair; it does not make the response odd. mean [even part]:  
moments 0.153; centers 0.387; intervals 0.333.

## F Bootstrap stability of $C_m$



Block 1, 95% multinomial bootstrap. Solid ellipse  $C_1$ , dashed  $C_2$ ; open centre marks a suppressed phase ( $R_m < 0.15$ , dotted circle, or ellipse covering the origin).